

CONNOR F. CRUZ

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Education

Yale University — New Haven, CT

Expected May 2030

B.S. candidate; prospective major in Biomedical Engineering

Charlotte Latin School — Charlotte, NC

June 2026

- GPA 4.31/4.33 unweighted; Class Rank: 1/143; National Merit Scholar; SAT 1600.

North Carolina School of Science and Mathematics (NCSSM Online)

June 2026

- GPA 4.00/4.00. College-level coursework in Computational Science, Cryptography, Nanotechnology.

Technical Projects

Na'dañña' Guam — MIT Fab Academy Final Project

2024

Technical Skills: Embedded C/Arduino, KiCad, Fusion 360; RP2040 + ATtiny1614

- Designed and fabricated an interactive map highlighting locations significant to Guam's history and culture through photographs and location-triggered digital displays.
- Developed the exhibit's embedded electronics, milled custom components including a custom PCB integrating a Seeed Xiao RP2040, an ATtiny1614, and SS49E hall-effect sensors concealed beneath the board; designed/authored the sensing and display firmware and user interface.
- Solved a half-inch magnetic detection limit by inlaying the acrylic map into a CNC-milled pocket rather than surface-mounting it, making marker placement register reliably.

E.M.I. (Engaged Memory Imaging) — Founder & Sole Developer

2025 – Present

Technical Skills: JavaScript, React; originally JavaFX + Firebase

- Designed and developed an interactive memory-support application to assist persons with dementia by converting photographs of familiar persons into low-friction recall exercises.
- Implemented three interaction modes: a concentration-style matching game, a browsable family gallery, and a multi-difficulty recall quiz. Rearchitected a stalled JavaFX prototype as a browser-based app.
- Conducted an accessibility review (large targets, high-contrast type, keyboard-navigable flows) focused on users with limited fine motor control.

Research & Publications

NCSSM Journal of Student Computational Science

December 2025

- Published: "Modeling Cultural Assimilation Using Global Migration, Economic Integration, and Cultural Diversity Metrics."
- Built and validated a computational model of assimilation dynamics from international migration data utilizing LaTeX for scientific and technical documentation.

Technical Skills

Applications: Microsoft Word, Excel, PowerPoint; Google Docs, Sheets, Slides; Onshape, KiCad, Fusion 360; Git/GitHub; Canva

Programming Languages: Python, Java, C++ (advanced); JavaScript, HTML/CSS, LaTeX

Tools: React, Next.js, Firebase, Arduino, Vercel

Fabrication: CNC milling, laser & vinyl cutting, 3D printing, PCB design and production, embedded systems

Methods: Computational modeling, data analysis, cryptography, accessible design

Selected Programs & Awards

- **Programs:** MIT Fab Academy Diploma Recipient (2024); PROMYS (2024); MathILy Early Research (2023)
- **Recognitions:** College Board National Indigenous Recognition Award (2024); Scholastic Writing Awards: Two Gold and Three Silver Keys for Writing (2024-2026)

Leadership & Service

Math Club — Leader, Charlotte Latin School

2024 – 2025

- Led weekly problem-solving sessions and organized preparation for mathematics competitions.
- Organized and staffed the Math Club information table at Charlotte Latin's Activities Fair, speaking with prospective members and encouraging student participation.

Mel's Diner & Meal Bags To Go — Volunteer, Charlotte, NC

2019 – 2026

- Served for six years with a church-based program preparing and distributing meals and pantry bags to food-insecure residents in the local community.